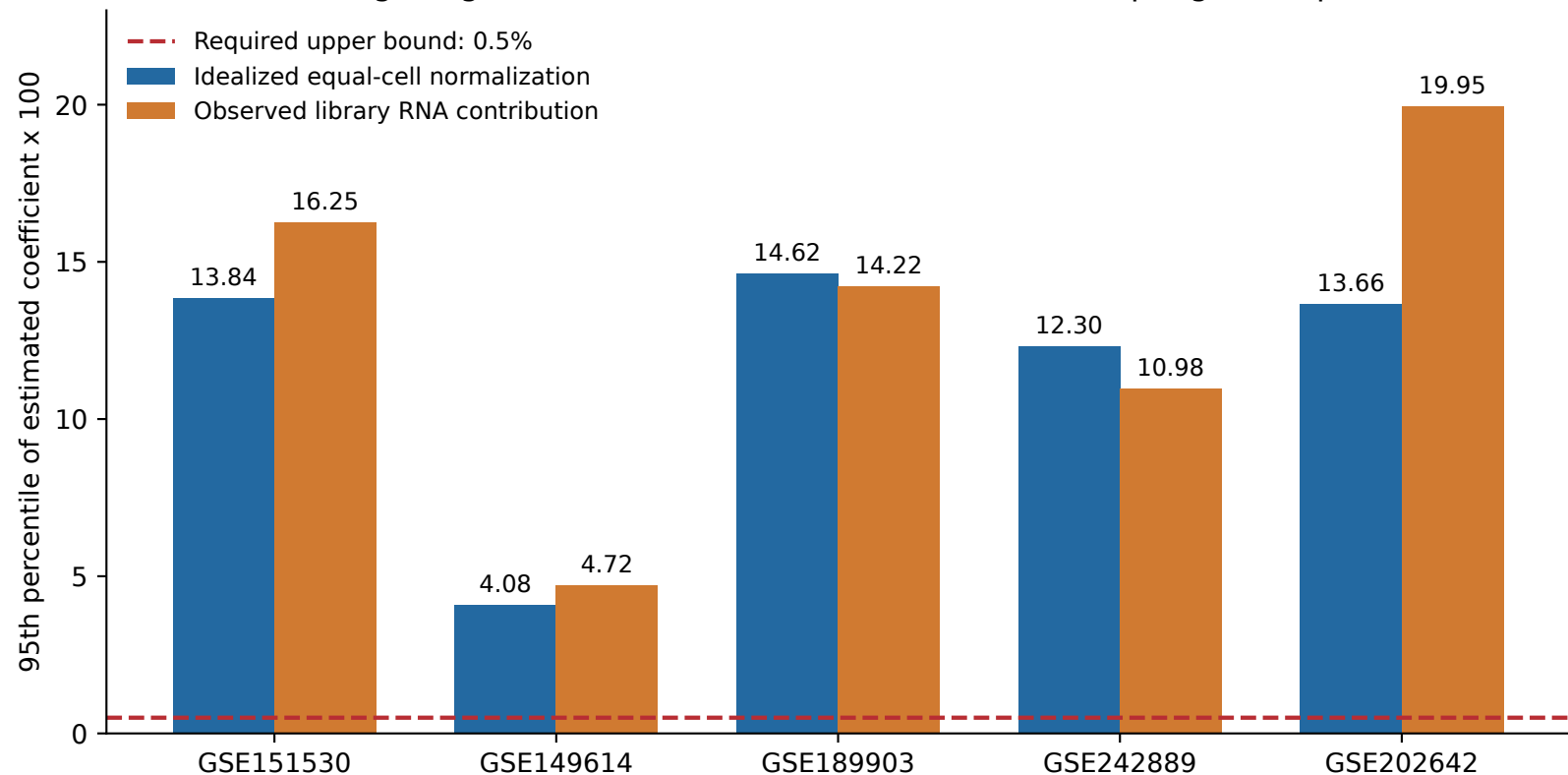


False target signal when no PGAM5 RNA-detected macrophages are present



All target-absent cases: standard zero-dose, macrophage-dose challenges, PGAM5-detected nonmacrophages, cycling nonmacrophages and pure competitors. Mixtures are resampled, not independent patients. Every cohort/unit exceeds the fixed 0.5% validation gate.